

Classification and FairML with the COMPAS dataset

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Abstract—This report presents a fair machine learning analysis of the COMPAS recidivism dataset. COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) was a risk assessment tool used to estimate the likelihood that a defendant would reoffend. The project develops classification models for predicting recidivism and evaluates them not only by conventional predictive performance metrics, but also by fairness metrics across racial groups. The analysis focuses on the binary target variable (*is_recid*), where 1 indicates recidivism and 0 indicates no recidivism. The protected attribute is race, with the main fairness comparison between African-American and Caucasian defendants. Five machine learning models are trained: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and Support Vector Machine. In addition, the original COMPAS score is evaluated as a baseline classifier. Fairness is evaluated using metrics such as Statistical Parity Difference, Disparate Impact, False Positive Rate Difference, False Negative Rate Difference, Equal Opportunity Difference, Average Odds Difference, and Selection Rate Difference. The AI Fairness 360 (AIF360) library is used for fairness measurement and for bias mitigation through the Reweighting preprocessing algorithm. The final model is selected based on the balance between accuracy, F1-score, ROC-AUC, and fairness gaps rather than accuracy alone.

Index Terms—COMPAS, Fair Machine Learning, AIF360, Classification, Reweighting, Recidivism, Fairness Metrics.

I. INTRODUCTION

Machine learning models are increasingly used in high-impact decision-making areas such as healthcare, finance, recruitment, and criminal justice. In such applications, model accuracy alone is not sufficient because predictions may affect different social groups unequally. Therefore, fair machine learning is important when datasets contain sensitive attributes such as race or gender.

This project uses the COMPAS dataset, which is related to recidivism risk assessment. COMPAS was used to estimate whether a defendant was likely to reoffend. The dataset became widely discussed after ProPublica reported racial differences in COMPAS prediction errors, especially between black and white defendants [1]. This shows that a model can be useful for prediction but still unfair if its errors are not equally distributed across protected groups.

The task of this project is to build classification models for predicting recidivism and evaluate them using both performance and fairness metrics. The target variable is recidivism, and race is used as the protected attribute. Several models are compared, including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and Support Vector Machine.

This work also uses AI Fairness 360 (AIF360), an open-source toolkit for detecting and mitigating algorithmic bias [2]. AIF360 is used to calculate fairness metrics and apply the Reweighting method, which assigns different weights to training samples based on protected group and label combinations [3]. The main goal is to select a model that gives a good balance between predictive performance and fairness.

II. SCIENTIFIC BACKGROUND

The prediction task in this study is binary classification. The target variable is *is_recid*, where 0 means that the person did not reoffend and 1 means that the person reoffended. Traditional evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC are useful for measuring prediction quality. However, these metrics describe overall performance and do not show whether errors are distributed equally across groups. Fair machine learning addresses this limitation by evaluating model behavior with respect to protected attributes. In this project, race is the protected attribute. African-American defendants are treated as the unprivileged group and Caucasian defendants as the privileged group, following the terminology used in fairness toolkits such as AIF360. A favorable outcome is defined as no recidivism, while an unfavorable outcome is recidivism. Several fairness metrics were used. Statistical Parity Difference compares favorable prediction rates between groups. Disparate Impact compares the ratio of favorable prediction rates. False Positive Rate Difference measures whether one group is more often wrongly classified as high risk. False Negative Rate Difference measures whether one group is more often wrongly classified as low risk. In the COMPAS context, false positives are especially important because a false positive means that a person who did not reoffend is predicted as high risk.

III. DATASET AND PREPROCESSING

The COMPAS dataset was obtained from ProPublica’s public analysis of recidivism risk assessment scores [1]. The original file contains defendant information, demographic variables, criminal history variables, COMPAS scores, and recidivism outcomes. The first preprocessing step was to remove rows with invalid target values. For binary classification, only records with is_recid equal to 0 or 1 were retained. The fairness analysis was focused on African-American and Caucasian defendants because these were the two largest groups and the main groups discussed in the COMPAS bias literature. Other racial groups were removed to make the group comparison clearer and more stable. A binary protected attribute called race_binary was created, where 0 represents African-American defendants and 1 represents Caucasian defendants. The variables age_cat and c_charge_degree are treated as ordinal features. The age category has a natural order and was mapped as Less than 25 = 0, 25 - 45 = 1, and Greater than 45 = 2. The charge degree variable contained detailed codes such as F1, F2, M1, MO3, NI0, CO3, CT, and X. These were first grouped by the leading alphabetic charge type, such as F, M, N, C, T, and X. Then the grouped charge type was ordinally encoded using a project-defined order. Other variables such as sex and race were one-hot encoded because they do not have a meaningful numerical order.

The finalized preprocessed dataset consists of 7,267 total records, distributed as 4,452 instances of no recidivism (61.27%) 2,815 instances of recidivism (38.73%). A baseline analysis of the protected attribute reveals a significant historical disparity: the favorable outcome rate (is_recid = 0) is 57.88% for African-American defendants, compared to 66.46% for Caucasian defendants. This baseline Statistical Parity Difference of -0.0858 confirms a structural bias in the underlying ground-truth labels, directly justifying the need for a pre-processing mitigation strategy like AIF360 Reweighing.

Final encoded dataset shape: (7267, 11)

age	juv_fel_count	juv_misd_count	juv_other_count	priors_count	is_recid	race_binary	age_cat.ordinal	c_charge_degree.ordinal	sex_Male	race_Caucasian
3	34	0	0	0	0	1	0	1	2	True
4	24	0	0	1	4	1	0	0	2	True
9	23	0	1	0	1	0	0	0	2	True
12	41	0	0	0	14	1	1	1	2	True
17	39	0	0	0	0	0	1	1	1	False

Fig. 1. Main preprocessing decisions.

IV. BASELINE COMPAS ANALYSIS

Before training new machine learning models, the original COMPAS score was evaluated as a baseline classifier. The score_text variable was converted into a binary prediction. Low risk was treated as predicted no recidivism, while Medium and High risk were treated as predicted recidivism. This baseline allowed the later machine learning models to be compared with the original COMPAS-style decision rule. The baseline analysis is independent of the later feature encoding decisions because it uses only the COMPAS score categories and the true recidivism label. Accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix values were calculated

as shown in Figure 3. In addition, Figure 2 shows group-based error rates that were computed separately for African-American and Caucasian defendants. This step was important because the COMPAS debate focuses not only on overall accuracy but also on unequal false positive and false negative rates.

Group	TN	FP	FN	TP	False Positive Rate	False Negative Rate	True Positive Rate	True Negative Rate	Selection Rate
0 African-American	1170	1378	519	1335	0.540816	0.279935	0.720065	0.459184	0.616311
1 Caucasian	1293	611	456	505	0.320903	0.474506	0.525494	0.679097	0.389529

Fig. 2. Fairness-related error rates for each racial group.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	FPR Difference	FNR Difference	TPR Difference	Selection Rate Difference
0 COMPAS Baseline	0.592129	0.480543	0.653641	0.553883	0.603438	0.219913	-0.19457	0.19457	0.226782

Fig. 3. COMPAS baseline summary.

V. MODELING APPROACH

The cleaned and encoded dataset was divided into training and testing subsets using an 80/20 split with stratification by the target variable. Stratification was used so that the proportion of recidivism and no recidivism remained similar in the training and test sets. Feature scaling was applied for Logistic Regression and Support Vector Machine because these models are sensitive to feature magnitude. Tree-based models were trained on unscaled features. Five classification models were implemented: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and Support Vector Machine using the scikit-learn machine learning library [4] as shown in Figure 4. Logistic Regression was used as an interpretable baseline. Decision Tree was included because it is easy to visualize and explain. Random Forest and Gradient Boosting were used as ensemble methods that can capture non-linear patterns in tabular data. Support Vector Machine was included as an additional non-linear classifier but was less interpretable and computationally more expensive.

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	Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
1	Logistic Regression	0.65433287482806	0.6107142857142858	0.3037300177619893	0.40569395017793597	0.6802134229606106
2	Decision Tree	0.6519944979767263	0.5959595959595959	0.31439721136767317	0.4116279086767442	0.6896527833695153
3	Random Forest	0.6430536451169189	0.5862650602409639	0.3339263996447602	0.4201117318438754	0.6889322872152749
4	Gradient Boosting	0.6499112242090785	0.5771428571428572	0.35879216472468916	0.4424972617743702	0.679383852336263
5	Support Vector Machine	0.6341127922971114	0.5551601423487544	0.27708703374777977	0.3996682464454976	0.6423820771964763

Fig. 4. Performance of the Five Classification Models.

VI. HYPERPARAMETER TUNING

Hyperparameter tuning was performed using GridSearchCV. The purpose was to avoid relying only on default model settings. For example, tree depth, number of estimators, learning rate, and regularization strength can strongly influence predictive performance and overfitting. Tuning was mainly applied to Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. The scoring metric for tuning was F1-score because the task involves balancing false positives

and false negatives. The performance metrics of the tuned models are presented in Fig. 5. A model with high accuracy but poor recall or poor precision may not be suitable for this type of classification task. Hyperparameter tuning improves predictive performance, but it does not automatically improve fairness. Therefore, fairness metrics were recalculated after model training and after bias mitigation.

	Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
0	Tuned Logistic Regression	0.6554	0.6107	0.3037	0.4057	0.6802
1	Tuned Decision Tree	0.6238	0.5244	0.3055	0.3861	0.6476
2	Tuned Random Forest	0.6403	0.5549	0.3588	0.4358	0.6430
3	Tuned Gradient Boosting	0.6513	0.5773	0.3712	0.4519	0.6768

Fig. 5. Hyperparameter Tuned Model Results.

VII. FAIRNESS EVALUATION AND BIAS MITIGATION

Fairness was evaluated using both AIF360 and manual group-based calculations. AIF360 was used to compute metrics such as Statistical Parity Difference, Disparate Impact, Equal Opportunity Difference, Average Odds Difference, and Theil Index. Manual calculations were also used for False Positive Rate Difference, False Negative Rate Difference, True Positive Rate Difference, and Selection Rate Difference because these values are easier to explain in the COMPAS context. The fairness evaluation results for all trained models are summarized in Table I.

TABLE I
FAIRNESS RESULTS FOR ORIGINAL MODELS.

Model	FPR Diff.	FNR Diff.	Sel. Diff.	DI	Avg. Odds
Log. Reg.	0.1112	-0.2486	0.1731	0.8104	-0.1799
Decision Tree	0.0813	-0.1947	0.1343	0.8470	-0.1380
Random Forest	0.1457	-0.2794	0.2056	0.7709	-0.2126
Grad. Boost.	0.1487	-0.2786	0.2084	0.7649	-0.2137
SVM	0.1279	-0.2310	0.1742	0.8092	-0.1795

Bias mitigation was implemented using the AIF360 Reweighting algorithm. Reweighting is a pre-processing method that modifies sample weights before model training [3]. It does not remove rows, change labels, or modify the test set. Instead, it assigns sample weights to training examples based on their protected group and outcome combination. Underrepresented or disadvantaged group-outcome combinations receive higher importance during model training, while overrepresented combinations receive lower importance. Mathematically, the Reweighting algorithm alters the dataset's empirical distribution by assigning higher weights to unprivileged favorable instances and lower weights to privileged favorable instances. This adjustment successfully nullifies the baseline selection gap of 0.0858 discovered during preprocessing, ensuring that the model is trained on a statistically balanced representation of both racial groups. In this project, Reweighting was applied to the training data only. The resulting instance weights were passed to models that support the `sample_weight` parameter, including Logistic Regression, Random Forest, and

Gradient Boosting. The models were then evaluated on the unchanged test data. The comparison between the original and reweighed model performance metrics is shown in Fig. 6. The fairness performance before and after applying Reweighting is presented in Fig. 7. This is important because fairness and performance must be tested on unseen data rather than on modified training data. Decision Tree and SVM are excluded from the Reweighting comparison because they do not support the `sample_weight` parameter used by AIF360's Reweighting-derived instance weights.

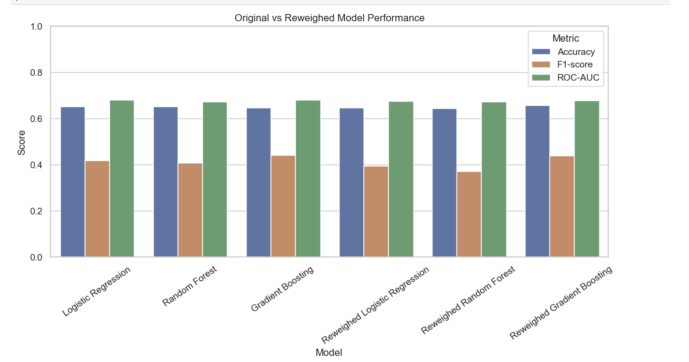


Fig. 6. Comparison of Original and Reweighed Models performance.

	Model	FPR Difference	FNR Difference	TPR Difference	TNR Difference	Selection Rate Difference	Training Method
0	Logistic Regression	0.1112	-0.2486	0.2486	-0.1112	0.1731	Original
1	Random Forest	0.1457	-0.2794	0.2794	-0.1457	0.2056	Original
2	Gradient Boosting	0.1487	-0.2786	0.2786	-0.1487	0.2084	Original
3	Logistic Regression	0.0060	-0.0300	0.0300	-0.0060	0.0247	Reweighed
4	Random Forest	0.0422	-0.1162	0.1162	-0.0422	0.0797	Reweighed
5	Gradient Boosting	0.0241	-0.0705	0.0705	-0.0241	0.0530	Reweighed

Fig. 7. Comparison of Original and Reweighed Fairness performance.

VIII. RESULTS AND DISCUSSION

The main result of the project is that predictive performance and fairness should be analyzed together. The original models achieved reasonable classification performance, but their fairness gaps were larger than the reweighed models. After applying AIF360 Reweighting, the points in the accuracy-fairness trade-off plot moved toward the left, meaning that the Fairness Gap Score decreased. The relationships between predictive performance and fairness are illustrated in Fig. 8, Fig. 9, and Fig. 10, showing accuracy vs. Fairness Gap Score, F1-score vs. Fairness Gap Score, and ROC-AUC vs. Fairness Gap Score respectively. The accuracy stayed relatively similar, showing that fairness was improved without a large loss of predictive performance. These figures illustrate the trade-off between predictive performance and fairness across models. Better-performing models lie toward the upper-left region, corresponding to higher accuracy (or F1-score/ROC-AUC) and lower Fairness Gap Scores. The F1-score-fairness plot captures the balance between precision and recall, while the ROC-AUC-fairness plot reflects class separability across thresholds. Overall, the results indicate that desirable models

achieve high predictive performance while maintaining a low fairness gap.

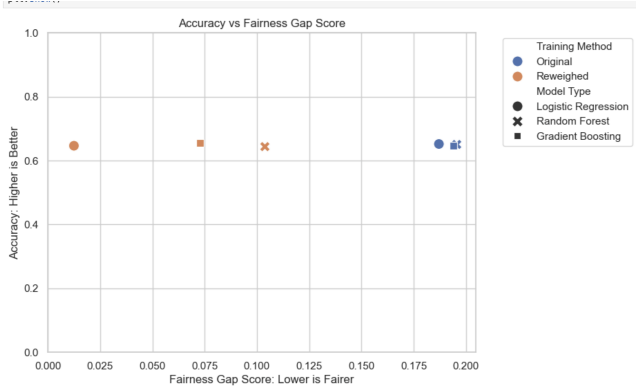


Fig. 8. Accuracy vs Fairness Gap Score.

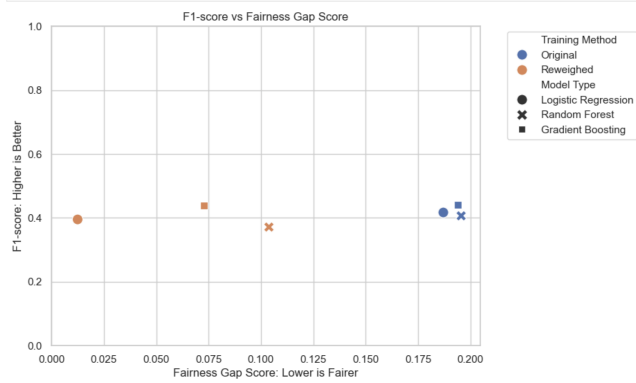


Fig. 9. F1-score vs Fairness Gap Score.

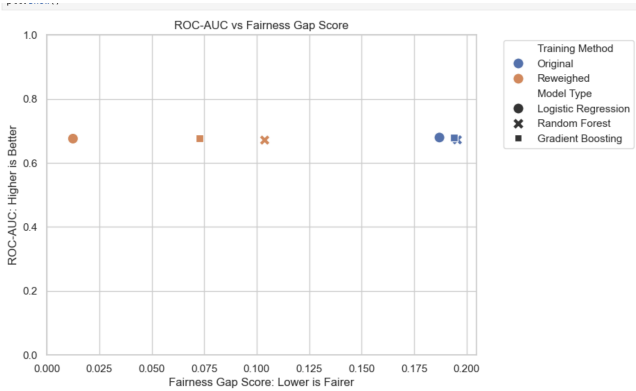


Fig. 10. ROC-AUC vs Fairness Gap Score.

The Fairness Gap Score was defined as the average of the absolute values of False Positive Rate Difference, False Negative Rate Difference, and Selection Rate Difference. It is not a universal fairness metric; it is a project-level summary score used to compare the tested models. A lower score means that group differences are smaller for the selected fairness criteria.

A. Final Fair Model Selection

The final model was selected by considering both predictive performance and fairness. The models were ranked using accuracy, F1-score, ROC-AUC, and Fairness Gap Score. The overall ranking of all models based on combined performance and fairness metrics is shown in Fig. 11. The Fairness Gap Score summarizes the differences in false positive rate, false negative rate, and selection rate between African-American and Caucasian defendants. A lower score means a fairer model.

	Model Type	Training Method	Accuracy	F1-score	ROC-AUC	FPR Difference	FNR Difference	Selection Rate Difference	Fairness Gap Score	Average Final Rank
0	Gradient Boosting	Reweighed	0.6534	0.4207	0.6769	0.0241	-0.0705	0.0530	0.0492	2.25
1	Logistic Regression	Original	0.6554	0.4057	0.6802	0.1112	-0.2486	0.1731	0.1776	2.50
2	Logistic Regression	Reweighed	0.6479	0.3710	0.6786	0.0060	-0.0300	0.0247	0.0202	3.25
3	Gradient Boosting	Original	0.6499	0.4425	0.6759	0.1487	-0.2786	0.2084	0.2119	3.50
4	Random Forest	Original	0.6431	0.4201	0.6689	0.1457	-0.2794	0.2056	0.2102	4.50
5	Random Forest	Reweighed	0.6424	0.4009	0.6679	0.0422	-0.1162	0.0797	0.0793	5.00

Fig. 11. Final Fair Model Selection.

The reweighed Gradient Boosting model achieved the best overall rank. It had an accuracy of 0.6534, F1-score of 0.4207, ROC-AUC of 0.6769, and Fairness Gap Score of 0.0492. Although the original Logistic Regression model had slightly higher accuracy and ROC-AUC, its fairness gap was much larger. Therefore, Gradient Boosting after reweighing was selected because it provided a better balance between accuracy and fairness.

	Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	FPR Difference	FNR Difference	Selection Rate Difference
0	COMPAS Baseline	0.5921	0.4805	0.6536	0.5539	0.6034	0.2199	-0.1946	0.2268
1	Gradient Boosting (Reweighed)	0.6534	0.5961	0.3250	0.4207	0.6769	0.0241	-0.0705	0.0530

Fig. 12. Compare Final Model with Original COMPAS Baseline.

The selected model was compared with the original COMPAS baseline, which is commonly used as a reference point in fairness studies of this dataset [1]. The selected model improved accuracy from 0.5921 to 0.6534 and ROC-AUC from 0.6034 to 0.6769. It also reduced the false positive rate difference from 0.2199 to 0.0241 and the selection rate difference from 0.2268 to 0.0530. The comparison between the selected Gradient Boosting model and the COMPAS baseline is shown in Fig. 12. This shows that the final model was not only more accurate than the COMPAS baseline, but also fairer across the two racial groups.

Overall, the reweighed Gradient Boosting model was selected as the final fair model for this project.

IX. CONCLUSION

This paper presented a practical FairML workflow for the COMPAS recidivism dataset. The study showed how to clean and preprocess the data, analyze the COMPAS baseline, train several classification models, evaluate fairness metrics, apply AIF360 Reweighting, and select a final model based on both accuracy and fairness. The updated preprocessing step used ordinal encoding for age category and charge degree, following the professor feedback.

The central conclusion is that the most accurate model is not necessarily the fairest model. In fairness-sensitive applications, group-based error rates must be considered together with predictive performance. AIF360 Reweighting was useful because it reduced fairness gaps while keeping model accuracy relatively stable. However, the results should be interpreted as an academic analysis only. A real criminal justice application would require deeper legal, ethical, and domain expert review.

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